Gosset-KEM: E8 Message Coding for ML-KEM and Correlation-Based Failure Boosting against Multi-Dimensional Lattice Codes

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Abstract

Replacing the one-bit-per-coefficient message encoding of ML-KEM (FIPS 203) by a multi-dimensional lattice code lowers the decryption failure rate (DFR); the margin can be spent on smaller ciphertexts or on more noise. We study the Gosset lattice E_8 in this role. We specify the encoding and decoding, derive the DFR by exact convolution with a union bound over the 240 Voronoi-relevant vectors of E_8, and validate the underlying independence assumption on 1.28–19.2 million blocks of real ring encryptions. We define *Compact* parameter sets (6–10% smaller ciphertexts, unchanged noise) and *MaxSec* parameter sets (FIPS 203 sizes, more noise). For MaxSec, the lattice-estimator reports +5 to +11 bits on the weakest of the key and ciphertext LWE instances. Our main contribution is a *correlation-based failure-boosting attack* that applies to any multi-dimensional message code. We prove that the conditional covariance between two decryption-error coefficients equals σ_S^2 times the negacyclic autocorrelation of attacker-known ciphertext randomness. A code that decodes along directions combining several coefficients is therefore sensitive to ciphertexts that an adversary selects by grinding messages; the standard 1-bit decoder is not. We give the exact form of this sensitivity, a tail model for it (validated against Monte Carlo within 15% down to 10^{-4}), and the resulting attack costs. With consecutive coefficient blocks, the cost of a first failure with at most 2^{64} queries drops to 2^{127} for the level-1 Compact set, below its claimed security. We prove that a coefficient layout in which the 28 pairwise index distances of each block are distinct reduces the worst-case sensitivity by a factor 2.24, and we exhibit such a layout. It raises the cost to 2^{192} while leaving sizes, speed and the average DFR unchanged. We also prove that no such layout exists for the Leech lattice. Finally, we report a negative result for E_8-structured *noise*, a constant-time and a first-order masked implementation, and compiler-induced timing leaks found during testing.

AI use designation (IACR AI Policy): Substantial AI contribution. *Author:* conceived the AFA-12/Gosset-KEM framework and its design, including the use of the E_8 lattice inside an ML-KEM-style scheme, from prior independent work (an earlier framework paper and an experimental harness). He also set the research direction, guided and decided each step, and ran the hardware timing tests. *AI system (Claude, Anthropic):* wrote the C implementation (including the masked version) and the analysis code. It also derived the proofs, found the correlation-based failure-boosting attack and the distinct-distance layout countermeasure, obtained the Leech impossibility result, ran the experiments and lattice-estimator evaluations, and drafted this paper. The author has reviewed the work and takes full responsibility for its content. All results can be reproduced from the published repository.

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1 Introduction

ML-KEM [1] encrypts a 256-bit message by mapping each bit to 0 or $\lceil q/2 \rfloor$ in one coefficient of a polynomial, and decrypts by rounding each coefficient. Lattice codes pack message points more densely in several dimensions, so a larger decryption noise is tolerated for the same failure probability [5, 4, 6, 3]. Liu and Sakzad recently proposed codes related to E_8 for Kyber [3].

Prior analyses compute the decryption failure rate of such schemes as an *average* over honestly generated ciphertexts. In the IND-CCA setting the adversary chooses which ciphertexts to submit, and failure-boosting attacks [7, 8] exploit ciphertexts whose failure probability is above average. The question we ask is whether the multi-dimensional structure of the code gives the adversary new ways to find such ciphertexts. It does, and the effect is large enough to matter at NIST level 1.

Contributions.

1. **Scheme and correctness** (§4–5). A complete specification of E_8 message coding inside ML-KEM (Algorithms 1–2), a proof that decoding errors are bounded by a union over the 240 relevant vectors (Lemma 3.1), exact DFRs for all three levels, and an empirical validation of the independence assumption (Table 2).
2. **Parameter sets and lattice security** (§6). Compact and MaxSec sets, evaluated with the lattice-estimator [10] on both LWE instances, and a search showing that MaxSec is optimal under FIPS-size and FIPS-DFR constraints.
3. **Correlation-based failure boosting** (§8). This is our main contribution. It consists of:

- the covariance identity (Lemma 8.1) and the variance-ratio formula (Lemma 8.2);
- the exact sensitivity of the consecutive layout and of distinct-distance layouts (Proposition 8.3);
- a diagonalisation of the tail problem (Lemma 8.5) with a Chernoff bound and a saddle-point estimate, both validated;
- attack costs, directional boosting after the first failure, and an impossibility result for the Leech lattice (Proposition 8.4).

4. **A negative result** (§7). E_8-structured noise loses $\log_2 D$ bits per block (Proposition 7.1), and real lattice-reduction attacks confirm it.

5. **Implementation** (§9). Constant-time decoding, a first-order masked decapsulation with a bitsliced masked E_8 decoder, and timing results on two compilers and two CPU vendors.

What is proven and what is measured. Lemmas and propositions are proven. Statements about the decryption failure rate at 2^{-140} and below, and about attack costs, rely on stated modelling assumptions (independence and Gaussian conditional models) that are standard in this literature [2, 7]. We validate each assumption where it is measurable and say so explicitly. Lattice-security figures are produced by the public lattice-estimator. All code and data are public (Appendix B).

2 Preliminaries

Notation. $q = 3329$, $n = 256$, $R_q = \mathbb{Z}_q[X]/(X^n + 1)$. For $x \in \mathbb{Z}_q$, $\mathrm{cent}(x)$ is its representative in $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]$. Vectors of polynomials are bold. β_η is the centred binomial distribution $\sum_{i=1}^{\eta}(a_i - b_i)$ with a_i, b_i uniform bits; it has variance $\eta/2$. $\mathrm{Compress}_d(x) = \lceil (2^d/q)x \rfloor \bmod 2^d$ and $\mathrm{Decompress}_d(y) = \lceil (q/2^d)y \rfloor$.

K-PKE (FIPS 203). Key generation samples $\mathbf{A} \in R_q^{k \times k}$, $\mathbf{s}, \mathbf{e} \leftarrow \beta_{\eta_1}^k$ and outputs $\mathbf{t} = \mathbf{A}\mathbf{s} + \mathbf{e}$. Encryption of m with coins derived from m samples $\mathbf{r} \leftarrow \beta_{\eta_1}^k$, $\mathbf{e}_1 \leftarrow \beta_{\eta_2}^k$, $e_2 \leftarrow \beta_{\eta_2}$, and outputs

$$\mathbf{u} = \mathrm{Compress}_{d_u}(\mathbf{A}^T\mathbf{r} + \mathbf{e}_1), \qquad v = \mathrm{Compress}_{d_v}(\mathbf{t}^T\mathbf{r} + e_2 + \mu),$$

where $\mu = \mathrm{Encode}(m) \in R_q$. Decryption computes $w = \mathrm{Decompress}_{d_v}(v) - \mathbf{s}^T\mathrm{Decompress}_{d_u}(\mathbf{u})$ and $m = \mathrm{Decode}(w)$. ML-KEM applies the Fujisaki–Okamoto transform with implicit rejection ($\mathrm{FO}^{\not\perp}$): the coins are $G(m \| H(\mathrm{ek}))$, decapsulation re-encrypts and compares, and the IND-CCA2 bound contains a correctness term δ [9].

Lemma 2.1 (Decryption error)**.** *Let* $\mathbf{c}_u = \mathrm{Decompress}_{d_u}(\mathbf{u}) - (\mathbf{A}^T\mathbf{r} + \mathbf{e}_1)$ *and* $c_v = \mathrm{Decompress}_{d_v}(v) - (\mathbf{t}^T\mathbf{r} + e_2 + \mu)$ *be the compression errors. Then* $w = \mu + \delta$ *with*

$$\delta = \mathbf{e}^T\mathbf{r} - \mathbf{s}^T(\mathbf{e}_1 + \mathbf{c}_u) + e_2 + c_v \in R_q.$$

Proof. $w = \mathbf{t}^T\mathbf{r} + e_2 + \mu + c_v - \mathbf{s}^T(\mathbf{A}^T\mathbf{r} + \mathbf{e}_1 + \mathbf{c}_u)$. Since $\mathbf{t}^T\mathbf{r} = \mathbf{s}^T\mathbf{A}^T\mathbf{r} + \mathbf{e}^T\mathbf{r}$, the terms in $\mathbf{A}$ cancel. □

Errors as inner products. For $a \in R_q$ with coefficient vector $(a_0, \ldots, a_{n-1})$, the product with b has coefficient i equal to $\sum_j a_j \,\mathrm{rot}_i(b)_j$, where $\mathrm{rot}_i(b)_j = b_{i-j}$ if $j \le i$ and $-b_{n+i-j}$ otherwise (negacyclic convolution). Define the secret $S = (\mathbf{e}, -\mathbf{s}) \in \mathbb{Z}^{2kn}$, the *ciphertext randomness* $C = (\mathbf{r}, \mathbf{e}_1 + \mathbf{c}_u) \in \mathbb{Z}^{2kn}$ and the offset $G = e_2 + c_v \in \mathbb{Z}^n$. Then

$$\delta_i = \langle S, C_i \rangle + G_i, \qquad C_i := \mathrm{rot}_i(C) \text{ (polynomial-wise)}. \tag{1}$$

C and G are deterministic functions of (m, ek), hence known to anyone who chooses m; S is the secret.

Algorithm 1 E8Encode(m), $m \in \{0,1\}^{256}$ as 32 bytes

for $j = 0$ to 31 **do**
  $b \leftarrow m[j]$; $c \leftarrow b_7$; $x_i \leftarrow b_i$ for $i = 0..6$; $x_7 \leftarrow x_0 \oplus \cdots \oplus x_6$
  for $i = 0$ to 7 **do**
    $\mu[L[8j+i]] \leftarrow 1665\, x_i$ if $c = 0$, else $832 + 1665\, x_i$ ▷ round($\frac{q}{2}(x_i + \frac{c}{2})$)
return μ

Algorithm 2 E8Decode(w), $w \in \mathbb{Z}_q^{256}$

for $j = 0$ to 31 **do**
  $Y_i \leftarrow 4 \cdot \mathrm{cent}(w[L[8j+i]])$ for $i = 0..7$; $U \leftarrow 2q$
  for $o \in \{0, q\}$ **do**
    $f_i \leftarrow \lfloor (Y_i - o)/U + \frac{1}{2} \rfloor$; $e_i \leftarrow Y_i - o - U f_i$
    if $\sum f_i$ odd **then**
      $t \leftarrow$ smallest index with $|e_t| = \max_i |e_i|$; $f_t \leftarrow f_t + \mathrm{sign}(e_t)$ ▷ sign(0) = +1
    $d_o \leftarrow \sum_i (Y_i - o - U f_i)^2$; $F_o \leftarrow f$
  $c \leftarrow [d_q < d_0]$; $m[j] \leftarrow c \cdot 2^7 + \sum_{i<7} (F_{cq,i} \bmod 2)\, 2^i$
return m

3 The Gosset lattice as a message code

$\mathsf{D}_8 = \{x \in \mathbb{Z}^8 : \sum x_i \equiv 0 \pmod 2\}$ and $\mathsf{E}_8 = \mathsf{D}_8 \cup (\mathsf{D}_8 + \frac{1}{2}\mathbf{1})$, where $\mathbf{1}$ is the all-ones vector. E_8 has determinant 1 and minimum squared norm 2. Its 240 minimal vectors are 112 of type $(\pm 1, \pm 1, 0^6)$ and 128 of type $(\pm\frac{1}{2})^8$ with an even number of minus signs. They are exactly its Voronoi-relevant vectors [12], and $2\mathbb{Z}^8 \subset \mathsf{D}_8 \subset \mathsf{E}_8$.

Code. The code is $\mathcal{C} = \frac{q}{2}\mathsf{E}_8 \bmod q\mathbb{Z}^8$. Since $q\mathbb{Z}^8 = \frac{q}{2} \cdot 2\mathbb{Z}^8 \subset \frac{q}{2}\mathsf{E}_8$, $\mathcal{C}$ is well defined, and $|\mathcal{C}| = [\mathsf{E}_8 : 2\mathbb{Z}^8] = 2^8$: one byte per 8 coefficients, the rate of ML-KEM. Coset representatives are $x \in \{0,1\}^8$ with $\sum x_i$ even, plus $c \cdot \frac{1}{2}\mathbf{1}$ with $c \in \{0,1\}$; this is 128 times 2, so 256 points.

Lemma 3.1 (Decoding region and union bound)**.** *Let $\hat{x}$ be a nearest point of the lattice $\Lambda = \frac{q}{2}\mathsf{E}_8$ to $p + \varepsilon$, where $p \in \Lambda$ and $\varepsilon \in \mathbb{R}^8$. If $\langle \varepsilon, \frac{q}{2}u \rangle < \frac{1}{2}\|\frac{q}{2}u\|^2$ for every relevant vector u, then $\hat{x} = p$. Consequently*

$$\Pr[\hat{x} \neq p] \;\le\; \sum_{u \in \mathcal{V}} \Pr\Big[\langle \varepsilon, u \rangle \;\ge\; \tfrac{q}{2}\Big],$$

where $\mathcal{V}$ is the set of the 240 minimal vectors. Decoding mod q is decoding in Λ, because $q\mathbb{Z}^8 \subset \Lambda$.

Proof. The Voronoi cell of 0 in Λ is $\{\varepsilon : \langle \varepsilon, v \rangle \le \frac{1}{2}\|v\|^2 \ \forall v \in \frac{q}{2}\mathcal{V}\}$, by the definition of relevant vectors. If ε lies in its interior, p is the unique nearest point. For $v = \frac{q}{2}u$ with $\|u\|^2 = 2$, the condition reads $\langle \varepsilon, u \rangle < \frac{q}{2}$. The bound follows by the union bound over the complement. □

Remark 3.2. Encoded coefficients are rounded to integers ($832.25 \to 832$, $1664.5 \to 1665$, $2496.75 \to 2497$). This adds a deterministic perturbation of at most 0.75 per coordinate, which we include in all computations. It is negligible compared with $q/2$.

4 Gosset-KEM

Gosset-KEM is ML-KEM with three changes: the message code of Algorithms 1 and 2, a fixed public coefficient layout L (a permutation of $\{0, \ldots, 255\}$, Appendix A), and the parameter sets of Table 1, which add $\eta = 4$ to the centred binomial distribution. Everything else is FIPS 203.

Table 1: Parameter sets. Sizes in bytes. DFR by Lemma 3.1 with exact convolution (§5).

Set	k	η_1	η_2	d_u	d_v	ek	ct	$\log_2$ DFR
ML-KEM-512	2	3	2	10	4	800	768	−139.1
ML-KEM-768	3	2	2	10	4	1184	1088	−165.3
ML-KEM-1024	4	2	2	11	5	1568	1568	−175.2
Gosset-KEM-512-Compact	2	3	2	9	4	800	704	−142.0
Gosset-KEM-768-Compact	3	2	2	9	5	1184	1024	−194.1
Gosset-KEM-1024-Compact	4	2	2	10	4	1568	1408	−215.0
Gosset-KEM-512-MaxSec	2	4	3	10	4	800	768	−146.6
Gosset-KEM-768-MaxSec	3	3	2	10	4	1184	1088	−166.8
Gosset-KEM-1024-MaxSec	4	3	2	11	5	1568	1568	−189.4

Proposition 4.1 (Correctness of decoding)**.** *Algorithm 2 returns, for each block, the coset of a nearest point of $\frac{q}{2}\mathsf{E}_8$ to $\frac{1}{4}Y$, with ties broken deterministically. In particular E8Decode(E8Encode(m))= m for every m.*

Proof. Scaling by 4 turns $\frac{q}{2}\mathsf{E}_8$ into $2q\mathsf{E}_8 = U\mathsf{E}_8$, which avoids fractions. For each coset $o/U \in \{0, \frac{1}{2}\mathbf{1}\}$ of D_8 in E_8, lines 4–7 are the Conway–Sloane nearest-point algorithm for D_8 [12]. Rounding each coordinate gives the nearest point of $\mathbb{Z}^8$; if its coordinate sum is odd, moving the worst-rounded coordinate one step towards the target gives the nearest point of D_8. Line 9 keeps the closer of the two candidates, and $\mathsf{E}_8 = \mathsf{D}_8 \cup (\mathsf{D}_8 + \frac{1}{2}\mathbf{1})$. The message bits are recovered because $f_i \bmod 2 = x_i$ for the transmitted point; x_7 is implied by the parity. For the round trip, the encoded point lies within 0.75 of a lattice point per coordinate (scaled by 4: ≤ 3), far inside the Voronoi cell, whose inradius is $\frac{q}{2} \cdot \frac{\sqrt{2}}{2} \cdot 4$. □

We verified Proposition 4.1 exhaustively for all 256 messages, and the constant-time C decoders agree with a reference decoder on $6.4 \cdot 10^6$ random noisy blocks (0 differences outside exact distance ties).

5 Decryption failure rate

Method. We compute the exact distribution of a single coefficient δ_i of Lemma 2.1:

- each product $e_j r_\ell$ and $s_j(e_{1,\ell} + c_{u,\ell})$ by enumerating the support;
- the kn-fold sums by repeated convolution;
- then $e_2 + c_v$ by convolution.

The compression errors are computed exactly from the rounding maps, assuming uniform inputs, as in the Kyber specification [2]. For the standard decoder this reproduces the published values exactly.

Assumption 5.1 (Block independence)**.** The 8 coefficients of δ that form a block are treated as independent draws from the marginal distribution.

Under Assumption 5.1, $\Pr[\langle \delta, u \rangle \geq q/2]$ follows by convolving two marginals (type $(\pm 1^2)$) or eight marginals scaled by $\frac{1}{2}$ (type $(\pm\frac{1}{2}^8)$). The per-message DFR is 32× the block bound of Lemma 3.1. At the ML-KEM parameters this gives $2^{-240.7}$, $2^{-279.4}$ and $2^{-335.0}$, against $2^{-139.1}$, $2^{-165.3}$ and $2^{-175.2}$ for standard decoding. A Gaussian approximation of δ_i gives $2^{-152.2}$, $2^{-162.7}$ and $2^{-290.9}$, consistent with the 2^{-286} reported in [3] for Kyber-1024. It is pessimistic because c_v is bounded and not Gaussian.

Table 2: Measured block failure rate in real encryptions (fresh keys per batch) against i.i.d. marginals and the union bound. The real/i.i.d. differences marked n.s. are not significant given the observed counts.

k	η	code	blocks	real	i.i.d.	union bound
2	12	standard	19.2 M	$4.995 \cdot 10^{-4}$	$4.984 \cdot 10^{-4}$	$4.939 \cdot 10^{-4}$
2	12	E_8	19.2 M	$1.875 \cdot 10^{-6}$ (36 ev.)	$1.550 \cdot 10^{-6}$ (n.s.)	$1.869 \cdot 10^{-6}$
2	13	standard	12.8 M	$1.559 \cdot 10^{-3}$	$1.549 \cdot 10^{-3}$	$1.551 \cdot 10^{-3}$
2	13	E_8	12.8 M	$1.711 \cdot 10^{-5}$ (219 ev.)	$1.600 \cdot 10^{-5}$ (n.s.)	$1.705 \cdot 10^{-5}$
2	16	E_8	1.28 M	$1.445 \cdot 10^{-3}$	$1.352 \cdot 10^{-3}$	$1.618 \cdot 10^{-3}$
3	16	E_8	1.28 M	$2.579 \cdot 10^{-2}$	$2.523 \cdot 10^{-2}$	$4.063 \cdot 10^{-2}$

Table 3: Weakest-link security ($\log_2$ of bit operations, lattice-estimator, full cost model). In each pair, the first value does not credit rounding noise and the second does. The cheapest attack is dual-hybrid in every case.

	512	768	1024
ML-KEM	136.0 / 139.7	196.4 / 196.4	262.3 / 262.3
Gosset-KEM MaxSec	141.6 / 144.9	201.2 / 207.2	267.6 / 271.7
gain	+5.6 / +5.2	+4.8 / +10.8	+5.3 / +9.4

Why independence is plausible, and where it fails. For a *random* ciphertext, $\mathbb{E}[\delta_a \delta_b] = \sigma_S^2 \, \mathbb{E}_C[A(b-a)] = 0$ (Lemma 8.1 below), so the coefficients are uncorrelated on average. They are not independent, since they share S. Section 8 shows that for *adversarially chosen* ciphertexts the conditional correlations are what matters.

Validation. We implemented the full ring PKE (any k, CBD, compression with $d_u = 10$, $d_v = 4$) and inflated η so that failures become observable. We compared the measured block failure rate with (a) i.i.d. coefficients drawn from the exact marginal and (b) the union bound. New keys were drawn for every batch. Table 2 shows the results: in the tail ($\eta = 12, 13$) the union bound matches the measured rate within 0.4%, and real/i.i.d. ratios are consistent with 1.

As for ML-KEM itself, the cryptographic DFRs (2^{-140} and below) extrapolate the validated model; they cannot be simulated directly.

6 Lattice security of the parameter sets

Instances. An attacker can target the key instance $(\mathbf{A}, \mathbf{t})$, with secret and error from β_{η_1}, or the ciphertext instance $(\mathbf{A}^T, \mathbf{u})$, with secret $\mathbf{r} \sim \beta_{\eta_1}$ and error $\mathbf{e}_1 \sim \beta_{\eta_2}$ plus the compression error $\mathbf{c}_u$. We report the weakest of the two, with and without crediting $\mathbf{c}_u$ as noise (modelled as a discrete Gaussian of the same variance). The Kyber team credits it for Kyber-512.

Tools. (i) A core-SVP implementation (primal uSVP 2016 estimate; dual attack with distinguishing advantage $4e^{-2\pi^2\tau^2}$ and sieve amortisation $2^{0.2075\beta}$). It reproduces the Kyber round-3 values: primal 118.6/181.9/254.9 against 118/182/256, and dual 117.0/180.2/252.6 against 117/181/253. (ii) The lattice-estimator [10] (commit f31b86f) with all attacks except Arora–GB and BKW, under its default cost model, which includes MATZOV's dual attack [11].

Compact sets have the noise of ML-KEM, and therefore its lattice security.

MaxSec is optimal under FIPS constraints. We searched $\eta_1 \in \{2, .., 5\}$, $\eta_2 \in \{2, 3, 4\}$, $d_u \in \{9, 10, 11\}$ and $d_v \in \{3, 4, 5\}$, requiring a ciphertext no larger than FIPS and a DFR no

larger than FIPS. Maximising the weakest-link core-SVP security, the MaxSec sets of Table 1 are the optimum at every level. Raising η_2 at levels 768 and 1024 would violate the DFR constraint.

7 A negative result: E_8-structured noise

Our starting point was to sample the LWE *error* blockwise from the integral copy $L = \sqrt{2}\mathsf{E}_8 = \{x \in \mathbb{Z}^8 : x \bmod 2 \in H_8\}$, where H_8 is the extended $[8, 4, 4]$ Hamming code and $\det L = 16$.

Proposition 7.1 (Entropy loss)**.** *Let $L \subseteq \mathbb{Z}^k$ be a sublattice of index D, and let X be supported on L. Let Y be the distribution on $\mathbb{Z}^k$ obtained by adding an independent uniform coset representative of $\mathbb{Z}^k/L$ to X. Then $H(Y) = H(X) + \log_2 D$. In particular, at equal per-coordinate variance and in the smooth regime, noise on L has $\log_2 D$ fewer bits of entropy per block than noise on $\mathbb{Z}^k$ of the same shape.*

Proof. The map $(x, c) \mapsto x + c$ from $L \times (\mathbb{Z}^k/L)$ to $\mathbb{Z}^k$ is a bijection, because cosets are disjoint and cover $\mathbb{Z}^k$. Hence $H(Y) = H(X) + H(\text{coset}) = H(X) + \log_2 D$. □

For E_8 this is 4 bits per block, half a bit per coordinate, which we confirmed numerically (3.986–4.000 bits at variances 0.5 to 9).

Attack. If the error lies in $L^{m/8}$ and $\gcd(D, q) = 1$, the primal embedding lattice Λ can be intersected with $\{x : x_{\text{err}} \in L^{m/8}\}$. The target $(\mathbf{e}, 1)$ is still in the intersection, and its determinant grows by $D^{m/8}$. In the uSVP success condition $\sigma\sqrt{\beta} \le \delta_\beta^{2\beta-d} \det^{1/d}$, this is equivalent to scaling σ by about $D^{-1/8}$, which is $1/\sqrt{2}$ for E_8. With the validated estimator, equal-variance E_8 noise lost 5, 11, 17 and 22 bits at $n = 256, 512, 768, 1024$ (equivalent σ ratio 0.704–0.707).

Experiment. We ran a Kannan-embedding attack with LLL ($n = 24$, $m = 48$, $q = 257$, 40 instances per point). At variance 12 per coordinate, a structure-aware attacker succeeded in 60% of instances on E_8 noise, the same as on i.i.d. noise at half the variance (60%), and against 0% on i.i.d. noise at the same variance. At variance 8 the figures were 95%, 92% and 48%. The weakness is invisible to an attacker who ignores the structure (30% at variance 8). Lattice structure belongs in the message, not in the noise.

8 Correlation-based failure boosting

8.1 Attacker model and cost

We use the model of [7]:

- The attacker knows ek and chooses messages m. Each m determines (C, G) through $G(m \| H(\text{ek}))$.
- The attacker submits ciphertexts to decapsulation and learns whether decryption failed. This is the standard assumption: a failure yields an unrelated shared key, observable at the protocol level.
- The attacker precomputes candidates and keeps those in a weak set $\mathcal{W}$.

Let $\alpha = \Pr_m[(C, G) \in \mathcal{W}]$ and let β be the failure probability of a ciphertext from $\mathcal{W}$ (over S). Obtaining one failure takes $Q = 1/\beta$ queries and $W = 1/(\alpha\beta)$ work. NIST bounds $Q \le 2^{64}$. The secret S is fixed but unknown, so we model it as drawn from its distribution.

8.2 Conditional covariance

For $C \in \mathbb{Z}^{2kn}$, define the negacyclic autocorrelation $A(d) = \sum_{\text{polys}} \sum_\ell C_\ell \, \tilde{C}_{\ell+d}$, where $\tilde{C}_{\ell+d} = C_{\ell+d}$ if $\ell + d < n$ and $-C_{\ell+d-n}$ otherwise.

Lemma 8.1 (Covariance)**.** *Let the coordinates of S be independent with mean 0 and variance σ_S^2, and independent of (C, G). Then for $0 \le a < b < n$:*

$$\mathrm{Cov}(\delta_a, \delta_b \mid C, G) = \sigma_S^2 \, A(b-a), \qquad \mathrm{Var}(\delta_a \mid C, G) = \sigma_S^2 \|C\|^2.$$

Moreover $A(n-d) = -A(d)$, so $A(n/2) = 0$.

Proof. By (1), $\mathrm{Cov}(\delta_a, \delta_b \mid C, G) = \sum_j \mathrm{Var}(S_j)(C_a)_j (C_b)_j = \sigma_S^2 \langle C_a, C_b \rangle$. Negacyclic rotation is an isometry and $\mathrm{rot}_b = \mathrm{rot}_{b-a} \circ \mathrm{rot}_a$, up to the sign convention. Hence $\langle C_a, C_b \rangle = \langle C, \mathrm{rot}_{b-a} C \rangle = A(b-a)$, which is independent of a; for $a = b$ it is $\|C\|^2$. For the symmetry, $\mathrm{rot}_{n-d} = -\mathrm{rot}_{-d}$ and $\langle C, \mathrm{rot}_{-d} C \rangle = \langle \mathrm{rot}_d C, C \rangle$, so $A(n-d) = -A(d)$. □

In ML-KEM, $\mathbf{s}$ and $\mathbf{e}$ both follow β_{η_1}, so $\sigma_S^2 = \eta_1/2$.

8.3 Sensitivity of a multi-dimensional decoder

A block uses positions $p_0 < \cdots < p_7$. By Lemma 3.1, it fails along a relevant vector u when $\langle \delta_{\text{block}}, u \rangle \ge q/2$.

Lemma 8.2 (Variance ratio)**.** *Let $T = \|C\|^2$ and $\rho_d = A(d)/T$. Then*

$$\mathrm{Var}(\langle \delta_{block}, u \rangle \mid C, G) = \sigma_S^2 \, T \, \|u\|^2 \, R(u), \qquad R(u) = 1 + \sum_{a<b} \frac{2u_a u_b}{\|u\|^2} \, \rho_{p_b - p_a}.$$

For the standard decoder ($u = e_i$), $R = 1$ identically.

Proof. $\mathrm{Var}(\sum_a u_a \delta_{p_a}) = \sum_{a,b} u_a u_b \mathrm{Cov}(\delta_{p_a}, \delta_{p_b})$. Apply Lemma 8.1 and separate the diagonal terms. □

So for a fixed norm T, the failure exponent along u scales by about $1/R(u)$. An attacker who grinds for ciphertexts with large $\max_u R(u)$ boosts failures; the standard decoder is immune. For a random C, ρ_d has mean 0 and standard deviation $s_\rho \approx (2kn)^{-1/2}$; we measured 0.022–0.035 across our sets.

Proposition 8.3 (Effect of the layout)**.** *For a type-$(\pm\frac{1}{2}^8)$ vector u, write $R(u) - 1 = \sum_\ell w_\ell \rho_\ell$ over distance classes $\ell \in \{1, \ldots, n/2 - 1\}$, where $w_\ell = \sum_{a<b} \epsilon_{ab} \frac{2u_a u_b}{\|u\|^2}$, summed over the pairs whose distance $p_b - p_a$ lies in class ℓ, and $\epsilon_{ab} = +1$ if $p_b - p_a = \ell$, -1 if $p_b - p_a = n - \ell$. Each term is $\pm\frac{1}{4}$. Write $\kappa = \max_u \left(\sum_\ell w_\ell^2 \right)^{1/2}$ for the worst-case multiplier of s_ρ, assuming the ρ_ℓ are uncorrelated.*

- *(a) With consecutive positions $p_a = 8j + a$, $\kappa = \frac{1}{4}\left(\sum_{d=1}^{7} (8-d)^2 \right)^{1/2} = \frac{\sqrt{140}}{4} \approx 2.958$, attained by the all-plus vector. The same distances occur in all 32 blocks, so one ground ciphertext boosts every block.*
- *(b) If the 28 distance classes $\min(p_b - p_a, \, n - (p_b - p_a))$ of every block are distinct, then $\kappa = \frac{\sqrt{28}}{4} \approx 1.323$. A given distance appears in at most one pair per block.*

For type-$(\pm 1^2)$ vectors, $R - 1 = \pm\rho_d$ for a single distance, so the multiplier is 1 in both layouts.

Proof. For type $(\pm\frac{1}{2}^8)$, $2u_a u_b / \|u\|^2 = \pm\frac{1}{4}$. Grouping the pairs by distance class gives w_ℓ; by Lemma 8.1, $\rho_{n-\ell} = -\rho_\ell$, so classes merge with a sign. (a) Distance d occurs in $8 - d$ pairs, and choosing all signs + makes every contribution add: $w_d = (8-d)/4$. (b) With distinct classes, each w_ℓ is a single term $\pm\frac{1}{4}$ over 28 classes. Uncorrelatedness is supported empirically: the mean correlation between different ρ_d was $-2 \cdot 10^{-4}$, with maximum $|\cdot|$ equal to 0.024. □

Appendix A gives a layout with property (b), found by randomised greedy search. Every distance class is used by at most 9 blocks. Property (b) can be checked directly from the table, and the reference implementation verifies it.

Proposition 8.4 (Leech lattice)**.** *For a code based on the Leech lattice Λ_{24} with blocks of 24 coefficients, no layout makes all pairwise distance classes of a block distinct.*

Proof. A block has $\binom{24}{2} = 276$ pairs. There are only $n/2 = 128$ distance classes, and class $n/2$ carries no correlation (Lemma 8.1). By the pigeonhole principle some class is shared by at least 3 pairs. □

Computing κ over the minimal-vector types of Λ_{24} ($(\pm 4^2)$, octads $(\pm 2^8)$, $(\mp 3, \pm 1^{23})$, scaled to norm 32) gives 4.90 for consecutive blocks and 2.14 for the best layout we found, against 1.32 for E_8. Higher-dimensional codes are therefore more exposed.

8.4 Tail of the sensitivity

The attack cost depends on $\Pr[R(u) \geq 1 + x]$ deep in the tail. Define the symmetric matrix $Q_u = \sum_\ell w_\ell (P^\ell + P^{-\ell})/2$, applied to each of the $2k$ polynomials, where P is the negacyclic shift. Then $R(u) - 1 = C^T Q_u C / C^T C$.

Lemma 8.5 (Diagonalisation)**.** *P is diagonalised by the vectors $(\omega_j^t)_{t<n}$ with $\omega_j = e^{i\pi(2j+1)/n}$, $j = 0, \ldots, n-1$. Hence Q_u has eigenvalues $\mu_j = \sum_\ell w_\ell \cos(\pi(2j+1)\ell/n)$ on each polynomial.*

Proof. $P^n = -I$, so the eigenvalues of P are the n-th roots of -1, namely the ω_j, with eigenvectors $(\omega_j^t)_t$ (the negacyclic DFT). $P^\ell + P^{-\ell}$ then has eigenvalue $\omega_j^\ell + \omega_j^{-\ell} = 2\cos(\pi(2j+1)\ell/n)$. □

Proposition 8.6 (Chernoff bound, Gaussian model)**.** *If the polynomials of C have independent $\mathcal{N}(0, \sigma_p^2)$ coordinates, then for $x > 0$:*

$$\Pr\big[R(u) - 1 \geq x\big] \;\leq\; \min_{\lambda > 0} \prod_p \prod_j \big(1 - 2\lambda\sigma_p^2(\mu_j - x)\big)^{-1/2},$$

over the λ for which all factors are positive.

Proof. The event is $\{C^T(Q_u - xI)C \geq 0\}$. Rotate each polynomial into the eigenbasis of Lemma 8.5; this is orthogonal over $\mathbb{R}$ when the conjugate pairs are combined. The form becomes $\sum_{p,j} \sigma_p^2(\mu_j - x) g_{pj}^2$ with independent standard normals g_{pj}. Apply Markov's inequality to $e^{\lambda Y}$ and use $\mathbb{E}[e^{\lambda a g^2}] = (1 - 2\lambda a)^{-1/2}$. □

The real entries (CBD plus rounding) are not Gaussian, so we treat Proposition 8.6 as a model and validate it. We also use the Lugannani–Rice saddlepoint approximation of the same quadratic form, which is more accurate. Against Monte Carlo with the true entries ($2 \cdot 10^5$ ciphertexts, Gosset-KEM-512-MaxSec), the results were:

- single direction, type $(\pm 1^2)$, at Monte Carlo levels $10^{-2}, 10^{-3}, 10^{-4}$: saddlepoint $1.02 \cdot 10^{-2}$, $0.97 \cdot 10^{-3}$, $0.97 \cdot 10^{-4}$; Chernoff $6.8 \cdot 10^{-2}$, $8.3 \cdot 10^{-3}$, $9.7 \cdot 10^{-4}$;
- single type-$(\pm\frac{1}{2}^8)$ direction of the distinct layout: saddlepoint $1.01 \cdot 10^{-2}$, $1.04 \cdot 10^{-3}$, $1.15 \cdot 10^{-4}$;
- union over all directions, $\Pr[\max R \geq r]$: saddlepoint $1.04 \cdot 10^{-2}$, $0.95 \cdot 10^{-3}$, $0.79 \cdot 10^{-4}$.

The tails are visibly heavier than Gaussian in ρ (at 3 and 4 standard deviations, 2–8 times the Gaussian value). A naive Gaussian model of ρ therefore underestimates the attack, while the Chernoff bound overestimates it.

Table 4: Correlation-based failure boosting ($\log_2$). Left: best per-ciphertext failure probability after grinding 2^{64} messages. Right: work for the first failure with $Q \le 2^{64}$. Norm-only grinding costs $\ge 2^{515}$ for every set, including ML-KEM.

	DFR after 2^{64} grinding			first-failure work		
Set	consecutive	distinct	ML-KEM	consecutive	distinct	claimed
512-Compact	−64.0	−84.1	−103.7	**127.4**	192.0	≈ 140
512-MaxSec	−70.9	−92.8	−103.7	147.6	232.4	144.9
768-Compact	−92.0	−119.9	−127.9	249.0	365.4	≈ 196
768-MaxSec	−91.5	−116.0	−127.9	242.2	386.6	207.2
1024-Compact	−124.3	−154.5	−141.3	554.4	790.7	≈ 262
1024-MaxSec	−108.8	−136.6	−141.3	336.4	552.3	271.7

8.5 Per-ciphertext failure model and validation

Given (C, G), we model δ_{block} as Gaussian with covariance $\sigma_S^2 M$ (Lemma 8.1) and mean G. The failure probability of a block is then bounded by the union over the 240 directions, each with variance from Lemma 8.2, averaged over the exact distribution of $\langle G, u\rangle$. We validated this per-ciphertext model on $8 \cdot 10^4$ real encryptions ($k = 2$, $\eta = 14$): 299 block failures observed against 303.1 predicted. Bucketed by $\max_u R(u)$, the counts were 162/147.4 (bottom 50%), 103/123.2 (50–90%), 30/28.9 (90–99%) and 4/3.6 (top 1%).

8.6 Attack cost

The attacker grinds on both T (the norm; $\Pr[T \ge t]$ by a Chernoff bound on the exact distribution of the squares) and R (saddlepoint). We treat the two as independent, since R is scale-free. We optimise W subject to $Q \le 2^{64}$, over thresholds t and x. With consecutive blocks, one ground ciphertext boosts the chosen direction in all 32 blocks (up to 14 directions per block); with the distinct layout, only the blocks that contain the chosen distance are affected.

With consecutive blocks, Gosset-KEM-512-Compact falls below its claimed security (Table 4); the MaxSec set at level 512 is at the boundary. With the distinct-distance layout every set exceeds its level by a wide margin. The layout changes neither sizes nor speed (a public gather/scatter). It does not change the average DFR either: in real encryptions at $\eta = 13$, both layouts gave $2.11 \cdot 10^{-5}$ over $3.84 \cdot 10^6$ blocks, and in an independent run the failing ciphertexts of the two layouts were disjoint.

8.7 After the first failure

Directional failure boosting [8] uses earlier failures to estimate the direction of S. We use its geometric model with attacker-favourable simplifications: the failing direction is known, and each failure contributes a vector with cosine $t = \theta/(\|S\|\|C\|\kappa_c)$ to S, where $\theta = q/4$ and $\kappa_c = 1$ for the standard decoder, and $\theta = q/2$ and $\kappa_c = \sqrt{2}$ for E_8. After n_f failures the estimate has cosine $\rho_{n_f} = \sqrt{n_f}\, t/\sqrt{1 + (n_f - 1)t^2}$. The attacker grinds for ciphertexts aligned with it, among 256 directions (standard) or $32 \cdot 240$ directions (E_8).

The work for the second failure, with $Q \le 2^{64}$, is $2^{84.7}$ for ML-KEM-512 against $2^{145.4}$ and $2^{147.9}$ for the 512 Compact and MaxSec sets. At level 768 it is $2^{103.0}$ against $2^{302.2}$ and $2^{200.2}$. Later failures become cheaper in every scheme, but remain costlier for Gosset-KEM than for ML-KEM. The first failure therefore dominates the attack, as for Kyber [8]. Multi-target attacks reduce the first-failure cost by at most the number of targets; they are not counted, as for ML-KEM.

9 Implementation and side channels

Correctness. The C99/AVX2 implementation passes all NIST ACVP vectors for ML-KEM (75 keyGen, 75 encapsulation, 30 decapsulation, and 60 key checks). An independent reference build, with a bit-level CBD and the Conway–Sloane decoder, agrees bit for bit on all nine sets. A deterministic digest over all sets, the hybrid modes and the DEM is identical with gcc and clang, on Linux and on Windows.

Constant time. The decoder is branch-free: rounding by comparisons, selections by masks, and value barriers. We used Welch's t-test, fixed against random inputs, with percentile cropping and inputs prepared in batches before timing; a positive control decoder contains secret-dependent branches.

- **Results.** On Intel Xeon and AMD Zen 3 (Ryzen 5 5600X), with gcc 13 and clang 18, all E_8 decoders gave $|t| \leq 1.8$ and the reference `poly_tomsg` $|t| \leq 3.6$ at 10^7 measurements; the positive control reached $|t| > 10^4$. Decapsulation of all sets gave $|t| \leq 3.5$ at 10^6 measurements, with one exception discussed below.
- **Compiler-inserted branches.** clang 18 `-O3` compiled the mask logic of a portable reference decoder into secret-dependent branches ($|t| \approx 1700$); gcc did not. gcc did the same to a Leech decoder. Value barriers fixed both.
- **A Zen 3 artefact.** One gcc binary showed $|t|$ of 10–19 for decapsulation, but only when the class was a *repeated* identical ciphertext. The effect was 5–10 cycles out of 85,000, it also appeared for ML-KEM-512, and valid against invalid random ciphertexts remained indistinguishable ($|t| \leq 1.9$). We attribute it to microarchitectural learning of repeated inputs; it carries no information about S.

Cost. On Zen 3 with gcc, the E_8 decoder (AVX2) takes 1,554 cycles, against 296 for the reference `poly_tomsg`. Decapsulation costs are:

- ML-KEM: 82,029, 125,615 and 181,041 cycles at levels 512, 768 and 1024;
- MaxSec: +3.7%, +3.0% and +2.6%, mostly from extra SHAKE output;
- Compact: within ±1.2% of ML-KEM.

Masking. We built a first-order masked decapsulation (Boolean and arithmetic sharing with fresh randomness per gadget):

- Keccak with ISW AND gadgets [13];
- the CBD on Boolean shares followed by B2A conversion mod q;
- an A2B conversion mod q by secure addition and masked conditional subtraction;
- the ciphertext comparison against public decompression intervals, followed by a masked AND reduction;
- the key selected on shares, so only the selected key is unmasked.

For the decoder we rewrote the E_8 decision without multiplications: with $a_i = \min(|Y_i|, 2q - |Y_i|)$, choose the half-integer coset iff $\sum a_i - 4q + p_A(2q - 2a_{\max}) - 2p_B a_{\min} > 0$, where p_A, p_B are the parities of the two candidates. This decision equals Algorithm 2 on $4 \cdot 10^5$ blocks, ties included, and allows a bitsliced masked decoder that processes 32 blocks per word. It takes 34k cycles, 12.3 times faster than per-block masking. Masked decapsulation costs 4.7–6.4 times the unmasked one.

Leakage assessment. The gadgets are individually first-order probing secure, but we do not give a formal composition proof. Our evidence is a simulated TVLA: Hamming weight of every intermediate share plus Gaussian noise ($\sigma = 1$), 43,956 points, up to $3 \cdot 10^4$ traces per class, four runs.

- First order: the maximum $|t|$ was 4.50–5.04. The Bonferroni threshold for 43,956 points is 4.87, and the single exceedance did not recur at the same point across independent runs.
- Second order: $|t| \approx 170$, as expected for first-order masking.

Hardware validation is open.

10 Limitations

- Gosset-KEM is not FIPS 203. A compliant deployment can use it only inside a hybrid that includes ML-KEM.
- DFRs below 10^{-6} and the attack costs rest on Assumption 5.1 and on Gaussian conditional models. These were validated where measurable, but the deepest tails are extrapolated, as for ML-KEM.
- The directional-boosting analysis uses a simplified geometric model.
- Key generation does not bind the parameter set: $G(d\|k)$ is identical for ML-KEM-512 and Gosset-KEM-512-Compact, so a key pair must never be reused across sets.
- The masking evidence is simulation only.

11 Conclusion

E_8 message coding buys margin in ML-KEM, either as 6–10% smaller ciphertexts or as +5 to +11 bits of lattice security at FIPS sizes. Multi-dimensional decoders, however, are sensitive to correlations between decryption-error coefficients, and those correlations are controlled by attacker-known ciphertext randomness. We characterised this sensitivity exactly. With consecutive blocks it is large enough to break the level-1 claim; a distinct-distance coefficient layout removes most of it at no cost. Higher-dimensional codes such as the Leech lattice are more exposed, and no layout can fully protect them. Any proposal of lattice codes for LWE-based KEMs should include this analysis.

AI use. This work is designated *substantial AI contribution* under the IACR AI Policy; the first page sets out the author's and the AI system's contributions. The author is responsible for the content, and all results are reproducible from the published repository.

A The layout table

Block j uses coefficients $L[8j], \ldots, L[8j+7]$:

```
0 18 58 159 162 197 227 228 | 27 42 78 112 133 145 209 235 | 43 47 54 74 97 190 231 246 | 96 142
194 204 217 218 226 243 | 21 37 61 84 161 166 203 244 | 32 71 77 79 138 157 223 252 | 30 36 73
124 168 184 195 216 | 1 45 59 101 104 129 186 255 | 9 10 29 44 72 105 193 248 | 16 57 60 70 82
180 189 245 | 39 65 103 114 153 155 242 249 | 11 86 128 134 232 237 241 253 | 13 122 143 151 158
182 196 251 | 3 20 38 48 126 139 222 254 | 12 25 31 34 81 91 111 115 | 26 55 89 106 167 225 229
234 | 40 56 63 99 107 132 137 207 | 17 19 98 108 201 202 214 233 | 8 15 35 52 120 169 221 224 |
53 64 117 118 152 177 179 191 | 46 92 136 172 187 192 220 250 | 14 24 66 100 123 131 149 208 | 6
22 51 119 154 156 160 212 | 85 140 164 176 183 205 230 247 | 2 62 83 94 181 185 200 238 | 5 50
```

113 121 174 175 210 213 | 7 76 102 116 148 206 215 236 | 23 69 87 93 109 110 165 198 | 67 125 135 144 146 170 188 219 | 4 49 80 150 163 173 178 211 | 68 75 90 95 130 141 239 240 | 28 33 41 88 127 147 171 199

B Reproducibility

All code, data and scripts are at https://github.com/calabroadrian/gosset-kem. `make test` runs the NIST vectors, the known-answer tests, the reference-versus-optimised and masked-versus-unmasked checks, and the cross-platform digest. Each result maps to a script in `research/`:

Result	Script
Proposition 7.1, §7 experiments	`v2/e8_construction_a.py`, `v2/estimator.py`, `v2/real_attack.py`
DFR (§5), Table 2	`v3/e8_message_coding.py`, `v3/correlation_check.py`
Table 3, MaxSec search	`v12/official_estimate.py`, `v12/coresvp.py`, `v12/rebalance.py`
Proposition 8.3, layout	`v13/layout_search.py`, `v13/corr_layout.py`
Tail models, Table 4	`v13/corr_chernoff.py`, `v14/corr_saddle.py`
Per-ciphertext validation	`v13/validate_corr.py`, `v13/dfr_layout_check.py`
Directional boosting, Leech	`v14/directional.py`, `v14/leech_layout.py`
Timing, masking	`bench/`, `research/v11/`, `research/local/`

References

[1] NIST. *FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard*. 2024.

[2] R. Avanzi, J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. M. Schanck, P. Schwabe, G. Seiler, D. Stehlé. *CRYSTALS-Kyber: Algorithm Specifications and Supporting Documentation*, version 3.02, 2021.

[3] S. Liu, A. Sakzad. *On the Maximum Toroidal Distance Code for Lattice-Based Public-Key Cryptography*. arXiv:2601.08452, 2026.

[4] S. Liu, A. Sakzad. *Lattice Codes for CRYSTALS-Kyber*. arXiv:2308.13981, 2023.

[5] C. Saliba, L. Luzzi, C. Ling. *Error Correction for FrodoKEM Using the Gosset Lattice*. arXiv:2110.01740, 2021.

[6] T. B. Paiva, M. A. Simplicio Jr, S. M. Hafiz, B. Yildiz, E. L. Cominetti, H. S. Ogawa. *Tailorable codes for lattice-based KEMs with applications to compact ML-KEM instantiations*. IACR TCHES 2025; ePrint 2024/1243.

[7] J.-P. D'Anvers, Q. Guo, T. Johansson, A. Nilsson, F. Vercauteren, I. Verbauwhede. *Decryption failure attacks on IND-CCA secure lattice-based schemes*. PKC 2019.

[8] J.-P. D'Anvers, M. Rossi, F. Virdia. *(One) failure is not an option: Bootstrapping the search for failures in lattice-based encryption schemes*. EUROCRYPT 2020; ePrint 2019/1399.

[9] D. Hofheinz, K. Hövelmanns, E. Kiltz. *A modular analysis of the Fujisaki–Okamoto transformation*. TCC 2017.

[10] M. R. Albrecht et al. *lattice-estimator*. https://github.com/malb/lattice-estimator, commit f31b86f.

[11] MATZOV. *Report on the Security of LWE: Improved Dual Lattice Attack*. 2022.

[12] J. H. Conway, N. J. A. Sloane. *Sphere Packings, Lattices and Groups*. Springer, 3rd ed., 1999.

[13] Y. Ishai, A. Sahai, D. Wagner. *Private circuits: securing hardware against probing attacks*. CRYPTO 2003.